

Some Popular Variable DC Power Supplies

	APLAB LAB Series - Digital Lab 30W-300W regulated DC power supplies	Agilent E36XX Series	Tektronix PWS4000 Series	BK Precision 1665 Series	Sciencetech 4074
					
Output voltages	0-16V, 0-32V, 0-64V and 0-128V DC	Different ranges between 0-120V	0-20V, 0-30V, 0-32V, 0-60V, 0-72V	1-20V, 1-40V and 1-60V	0-30V
Continuous current	—	—	—	—	1A
Output power	30 watts to 300 watts	30 watts to 60 watts	86 watts to 150 watts	200 watts	—
Metering	3-digit DPM ± 3 counts	3-digit DPM $\pm 0.5\% + 2$ counts	Vacuum fluorescent display	3 $\frac{1}{2}$ -digit LED, 1%+2 counts	LED indication for voltage and current (3-digit display for voltage and 3 $\frac{1}{2}$ -digit display for current), $\pm (1\% + 1 \text{ digit})$
Regulation	Constant-voltage mode Line regulation: $\pm 0.01\% \pm 2 \text{ mV}$ for $\pm 10\%$ change in load Load regulation: $\pm 0.01\% \pm 2 \text{ mV}$ for load change from zero to full load Constant-current mode Line regulation: $\pm 0.1\% \pm 250 \mu\text{A}$ for 10% line change Load regulation: $\pm 0.1\% \pm 250 \mu\text{A}$ for change in output from 0 to maximum	Constant-voltage mode Line regulation: $\pm 0.01\% \pm 2 \text{ mV}$ for $\pm 10\%$ change in load Load regulation: $\pm 0.01\% \pm 2 \text{ mV}$ for load change from zero to full load	Constant-voltage mode Line regulation: $\pm 0.01\% \pm 2 \text{ mV}$ Load regulation: $\pm 0.01\% \pm 1 \text{ mV}$ Constant-current mode Line regulation: $\pm 0.05\% \pm 0.1 \text{ mA}$ Load regulation: $\pm 0.05\% \pm 0.1 \text{ mA}$	Line regulation: 20 mV Load regulation: maximum 0.5%+200 mV	Line regulation: $\pm (0.05\% + 10 \text{ mV})$ Load regulation: $\pm (0.05\% + 10 \text{ mV})$
Ripple and noise	0.04% rms	200-350 μVrms	1 mVrms	20 mV	$\leq 1 \text{ mVrms}$
Transient response	100 μs	<50 μs	<400 μs	—	—
Stability	$\pm 0.2\%$ plus 5 mV in constant-voltage mode $\pm 0.5\%$ plus 5 mA in constant-current mode	—	$\pm 0.2\%$ plus 5 mV in constant-voltage mode $\pm 0.5\%$ plus 5 mA in constant-current mode	—	2.5 mV at 30V/1A, 5V/1A and $\pm 15\text{V/1A}$
Programming	No	No	Yes	No	No

real value to the specified value. Power supplies are becoming more and more digital these days. The embedded system of the power supply receives digital inputs from the user and converts them into analogue quantities using a digital-to-analogue converter. The analogue quantities are then used as reference for analogue regulators to produce the required output. Also, the output is fed back to an internal meter for measuring its value. The quality of this process determines the accuracy of the power supply.

In a datasheet, there are two specifications under accuracy: output accuracy and meter accuracy.

Output accuracy. Output accuracy defines the closeness of regulated output to the theoretically calculated value. It is represented as $\pm (\% \text{ theoretical value} + \text{offset value})$. So if you have set the knob at 5V, probably you are getting only 4.95V at the output.

Meter accuracy. It defines how close internally read values are to theoretically calculated output values. It is also generally represented as $\pm (\% \text{ theoretical value} + \text{offset value})$.

Resolution

Resolution is the minimum step change that you can select with the power supply. In low-current appli-

cations, resolution plays a significant role. Similar to accuracy, specifications under resolution are output resolution and meter resolution.

Output resolution. It is the minimum change in current or voltage that can be achieved through the power supply. Generally, 1mV/0.1mA resolution is used for lab experiments.

Meter resolution. Meter resolution is the minimum change that the power supply can read back and indicate on the display.

Regulation

Regulation is specified as load regulation and line regulation.